A Variational Control Framework for Optimal Portfolio Rebalancing Under Transaction Costs

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Abstract

We study portfolio rebalancing between one risky and one risk-free asset as a finite-horizon stochastic control problem with proportional transaction costs and constant-relative-risk-aversion (CRRA) preferences. Exploiting the homogeneity of CRRA utility we reduce the value function to a one-dimensional object in the risky-allocation fraction and solve the Bellman recursion exactly (to quadrature and grid resolution) by backward induction. The solution reproduces the classical *no-trade region*: a band of allocations around the frictionless Merton target inside which it is optimal not to trade, and toward whose boundary the investor trades when outside. We confirm the textbook comparative statics — the band widens with the transaction-cost rate (consistent with the small-cost $tc^{1/3}$ scaling) and narrows with risk aversion — and the horizon effect by which the band flares open near the terminal date.

We then ask a computational question: can a lightweight neural network imitate this optimal control? Training a 4,417-parameter multilayer perceptron by supervised regression on the dynamic-programming policy attains a validation mean-squared error of 2.1×10^{-4}, yet a paired Monte-Carlo evaluation (common random numbers, 20,000 paths) shows the learned policy is the *worst* performer by certainty equivalent. The mechanism is structural: a smooth regressor cannot represent the exact "do nothing" identity on a positive-measure set, so its tiny action-space errors inside the no-trade region are amplified into continuous spurious turnover (mean turnover 0.21 versus 3×10^{-5} for the optimal policy). We then show the diagnosis is actionable: restoring the inaction region — by a dead-zone wrapper at inference, or by penalizing excess turnover during training — closes the entire certainty-equivalent gap, leaving the repaired network statistically indistinguishable from the optimal control ($t = -0.4$, against $t = 20.1$ before the repair). Recovery sets in precisely as the dead-zone width crosses the network's in-band residual, confirming the mechanism. Both the failure and its repair are robust: across 10 seeds and 7 architectures spanning a factor of 198 in capacity, every student over-trades and underperforms, the largest network has the *worst* in-band residual, and the dead-zone recovers the optimum in all 17 cases. We conclude that imitating optimal *control* requires encoding the inaction region rather than minimizing action-space error. All results are reproducible from the accompanying code. **Keywords:** portfolio rebalancing; transaction costs; no-trade region; stochastic control; dynamic programming; imitation learning; reproducible research.

JEL classification: G11; C61; C63; C45.

1 Introduction

1.1 Motivation

A long-only investor who wants to hold a fixed fraction of wealth in a risky asset faces a basic tension. Random returns continuously perturb the realized allocation away from the target, but every corrective trade costs money. In a frictionless world the resolution is trivial: rebalance continuously to the Merton [1969] target. With transaction costs, continuous rebalancing is infinitely expensive, and the optimal policy instead tolerates deviations, trading only when the allocation drifts too far. Characterizing "too far" is the heart of the problem.

1.2 Transaction costs and rebalancing

Proportional transaction costs transform the single optimal allocation point into an interval of inaction — the *no-trade region*. This qualitative feature, first identified by Magill and Constantinides [1976] and rigorously established by Constantinides [1986], Davis and Norman [1990] and Shreve and Soner [1994], is robust across model specifications and is the single most important practical consequence of trading frictions. Its width encodes the cost–benefit trade-off of rebalancing and directly governs realized turnover.

1.3 Research questions

This paper is deliberately *reproductive and computational* rather than model-innovative. We pursue three questions.

Q1. Does a transparent, discrete-time dynamic program recover the classical no-trade region and its comparative statics?

Q2. Can a small neural network, trained by supervised imitation of the optimal policy, serve as a fast approximate controller?

Q3. How do the optimal policy, its learned approximation, and simple heuristics (periodic rebalancing, buy-and-hold) compare on economic performance, turnover, and cost?

Our contributions are (i) a clean, fully reproducible discrete-time solver that exploits CRRA homogeneity for a one-dimensional state; (ii) a closed-form proportional-cost wealth-update identity that makes the recursion exact and vectorizable; and (iii) the finding that naive action-space imitation *systematically fails* by violating the no-trade region, with a clear mechanistic explanation and statistically rigorous paired evidence.

2 Literature Review

Portfolio optimization. Merton [1969, 1971] solved the frictionless dynamic consumption–investment problem, giving the constant optimal risky weight $w^* = (\mu - r)/(\gamma\sigma^2)$ that anchors our no-trade region.

Transaction-cost models. Magill and Constantinides [1976] introduced proportional costs and the no-trade region concept; Constantinides [1986] quantified its asset-pricing implications; Davis and Norman [1990] and Shreve and Soner [1994] provided the singular-control and viscosity-solution foundations, including the small-cost asymptotics in which the half-width of the band scales like

$tc^{1/3}$. Dumas and Luciano [1991] obtained an exact infinite-horizon solution, and Kallsen and Muhle-Karbe [2010] developed the shadow-price approach. Liu [2004] extended the analysis to multiple risky assets and Gârleanu and Pedersen [2013] to predictable returns under quadratic costs.

Dynamic programming. The backward-induction machinery we use is standard [Bertsekas, 2017]; our only specialization is the homogeneity reduction that collapses the state to the allocation fraction.

Learning-based portfolio control. Machine learning has been applied to high-dimensional control via deep BSDE/PDE solvers [Han et al., 2018] and to hedging under frictions via deep hedging [Buehler et al., 2019]. We instead study the simplest learning paradigm — supervised *imitation* of a known optimal controller [Ross et al., 2011, Sutton and Barto, 2018] — precisely to isolate when and why it fails. Our negative result complements the deep-hedging literature by showing that the inaction structure induced by transaction costs is exactly what naive imitation cannot represent.

3 Mathematical Framework

3.1 Market model

Time is discrete, $t = 0, 1, \ldots, N$, with step length $dt = T/N$. The risky asset price S_t follows a discrete-time geometric Brownian motion, so its one-step gross return is

$$R_{t+1} = \frac{S_{t+1}}{S_t} = \exp\Big((\mu - \tfrac{1}{2}\sigma^2)\,dt + \sigma\sqrt{dt}\,Z_{t+1}\Big), \qquad Z_{t+1} \overset{\text{iid}}{\sim} N(0,1), \tag{1}$$

with drift μ and volatility σ. The risk-free asset grows deterministically by $R_f = e^{r\,dt}$. Equation (1) implies the lognormal moments $\mathbb{E}[R] = e^{\mu\,dt}$ and $\mathbb{E}[R^2] = e^{2\mu\,dt+\sigma^2 dt}$, which we use to validate the quadrature in Section 4.

3.2 Wealth, state, and control

At date t, before trading, the investor holds A_t dollars in the risky asset and C_t in cash, with total wealth $W_t = A_t + C_t$ and risky fraction $x_t = A_t/W_t$. The control is the *post-trade* risky fraction $y_t \in [0, 1]$ (no shorting, no leverage). The pair (W_t, x_t) is the state and y_t the action.

3.3 Transaction costs and the wealth-update identity

Normalize $W_t = 1$ (justified below). Pre-trade holdings are $A = x$ in risky and $1 - x$ in cash. Choosing post-trade fraction y requires trading δ dollars into the risky asset; the proportional cost $tc\,|\delta|$ is paid from cash, so post-trade wealth is $m = 1 - tc\,|\delta|$ and the post-trade risky holding is $y\,m = A + \delta$, i.e. $\delta = ym - x$. Substituting $m = 1 - tc|\delta|$ and solving the resulting fixed point yields a closed form.

Proposition 1 (Proportional-cost wealth multiplier)**.** *For $tc \in [0, 1)$ and $x, y \in [0, 1]$, the post-trade wealth multiplier is*

$$m(x, y) = \begin{cases} \dfrac{1 + tc\,x}{1 + tc\,y}, & y \geq x \quad (\textit{buying}), \\[2ex] \dfrac{1 - tc\,x}{1 - tc\,y}, & y < x \quad (\textit{selling}), \end{cases} \tag{2}$$

and the traded amount satisfies $\delta = (y - x)/(1 \pm tc\,y)$, so $\operatorname{sign}(\delta) = \operatorname{sign}(y - x)$ and $m(x, x) = 1$.

Proof. If $\delta > 0$ (buying), $m = 1 - tc\,\delta$ and $\delta = ym - x$. Then $m = 1 - tc(ym - x) \Rightarrow m(1 + tc\,y) = 1 + tc\,x$, giving the first branch. Substituting back, $\delta = ym - x = (y - x)/(1 + tc\,y) > 0$ iff $y > x$, confirming the case split. The selling branch ($\delta < 0$) is symmetric with $m = 1 + tc\,\delta$, yielding $m(1 - tc\,y) = 1 - tc\,x$ and $\delta = (y - x)/(1 - tc\,y)$. At $y = x$ both branches give $m = 1$ and $\delta = 0$. □

Equation (2) is exact (no linearization) and renders the entire recursion a handful of vectorized array operations.

3.4 Preferences and the Bellman recursion

The investor maximizes expected terminal CRRA utility

$$U(W) = \frac{W^{1-\gamma}}{1-\gamma}, \qquad \gamma > 0,\ \gamma \neq 1, \tag{3}$$

with relative risk aversion γ. After trading to fraction y at normalized wealth m, one market step (1) sends wealth to $W' = m\,\phi(y, R)$ and the fraction to $x'(y, R)$, where

$$\phi(y, R) = yR + (1 - y)R_f, \qquad x'(y, R) = \frac{yR}{\phi(y, R)}. \tag{4}$$

The value function $V_t(W, x) = \sup \mathbb{E}[U(W_N) \mid W_t = W, x_t = x]$ obeys $V_N(W, x) = U(W)$ and

$$V_t(W, x) = \max_{y \in [0,1]} \mathbb{E}_R\big[V_{t+1}(W', x')\big]. \tag{5}$$

3.5 Dimensionality reduction by homogeneity

Proposition 2 (Homogeneity reduction)**.** *There exist functions $g_t : [0, 1] \to \mathbb{R}$ with $V_t(W, x) = W^{1-\gamma}\, g_t(x)$ for all t. They satisfy the one-dimensional recursion*

$$\boxed{g_t(x) = \max_{y \in [0,1]} m(x, y)^{1-\gamma}\, \mathbb{E}_R\big[\phi(y, R)^{1-\gamma}\, g_{t+1}\big(x'(y, R)\big)\big], \qquad g_N(x) \equiv \frac{1}{1-\gamma}.} \tag{6}$$

Proof. By backward induction. At $t = N$, $V_N(W, x) = W^{1-\gamma}/(1 - \gamma)$, so $g_N \equiv 1/(1 - \gamma)$. Assume $V_{t+1}(W, x) = W^{1-\gamma} g_{t+1}(x)$. From Section 3.3, trading from (W, x) to fraction y gives post-trade wealth $W\,m(x, y)$, and the market step (4) sends wealth to $W' = W\,m(x, y)\,\phi(y, R)$ with next fraction $x'(y, R)$ independent of W. Then

$$V_t(W, x) = \max_y \mathbb{E}_R\big[(W')^{1-\gamma} g_{t+1}(x')\big] = W^{1-\gamma} \max_y m(x, y)^{1-\gamma} \mathbb{E}_R\big[\phi(y, R)^{1-\gamma} g_{t+1}(x')\big],$$

which is $W^{1-\gamma} g_t(x)$ with g_t as in (6). Both m and x' are wealth-independent, so the maximizer $y^*(t, x)$ depends only on (t, x). □

Proposition 2 is the workhorse: it removes wealth from the state space, leaving the scalar allocation fraction x as the only continuous state.

3.6 The no-trade region

The optimal policy is $y^*(t,x) = \arg\max_y(\cdot)$ in (6). Define the date-t *no-trade region*

$$\mathrm{NT}_t = \{x : \; y^*(t,x) = x\}. \tag{7}$$

The classical structural result — which our numerics confirm — is that $\mathrm{NT}_t = [\,\underline{x}_t,\ \overline{x}_t\,]$ is an interval around the Merton target w^*, with

$$y^*(t,x) = \begin{cases} \underline{x}_t, & x < \underline{x}_t \quad \text{(buy to lower edge)}, \\ x, & x \in [\underline{x}_t, \overline{x}_t] \quad \text{(hold)}, \\ \overline{x}_t, & x > \overline{x}_t \quad \text{(sell to upper edge)}. \end{cases} \tag{8}$$

Trading is "bang-bang to the boundary": when action is warranted, the investor moves exactly to the nearest edge of the inaction band, never overshooting into its interior.

4 Numerical Solution

4.1 State and action discretization

We discretize the fraction x on a uniform grid of $n_x = 201$ points in $[0,1]$; the action y shares this grid. The risky-return expectation in (6) is evaluated by $n_q = 21$-node Gauss–Hermite quadrature: for $Z \sim N(0,1)$, $\mathbb{E}[f(R)] \approx \pi^{-1/2} \sum_i w_i \, f\big(R(\sqrt{2}\, z_i)\big)$ with physicists' nodes (z_i, w_i). This reproduces the lognormal moments of Section 3 to machine precision (relative error $< 10^{-8}$ for $\mathbb{E}[R]$ and $\mathbb{E}[R^2]$).

4.2 Backward induction and policy extraction

Starting from $g_N \equiv 1/(1-\gamma)$ we iterate (6) backward. For each action y on the grid we precompute the next fractions $x'(y,R)$ and gross returns $\phi(y,R)$ over the quadrature nodes; the continuation value g_{t+1} is evaluated at $x'(y,R)$ by linear interpolation. The cost multiplier $m(x,y)$ from (2) is a full $(n_x \times n_y)$ matrix, and the objective $J_t(x,y) = m(x,y)^{1-\gamma}\mathbb{E}_R[\cdots]$ is maximized over y by a vectorized arg max. The per-step cost is $O(n_x\, n_y\, n_q)$ and the full $N = 52$-step solve completes in $\approx 0.02\,$s. The no-trade boundaries $\underline{x}_t, \overline{x}_t$ are read off as the range of x for which $|y^*(t,x) - x|$ is within one grid spacing.

The reduced value function. Figure 1 plots the solved $g_t(x)$ at three dates and makes the geometry of the problem visible. At every date g_t is concave and peaks at the Merton fraction $w^* = 0.300$, confirming that the frictionless target remains the centre of gravity once costs are added. Two features matter for what follows. First, the level of g_t rises with time-to-go (from $g_{51}(w^*) = -0.2494$ to $g_0(w^*) = -0.2227$, against the terminal value $g_N \equiv 1/(1-\gamma) = -0.25$): a longer horizon is worth more because the risky premium has longer to compound. Second, and more importantly, the *curvature* collapses near the horizon — the spread of g_t across the whole allocation interval falls from 6.1×10^{-3} at $t = 0$ to 9.4×10^{-4} at $t = 51$. A nearly flat value function means the allocation barely matters, so almost no deviation is worth a trade. This is the mechanism behind the horizon effect reported in Section 6: the band flares open precisely where g_t goes flat.

Base-case parameters. Table 1 lists the parameters used throughout. They place the Merton target $w^* = (\mu - r)/(\gamma\sigma^2) = 0.30$ in the interior of $[0,1]$, producing a clean band.

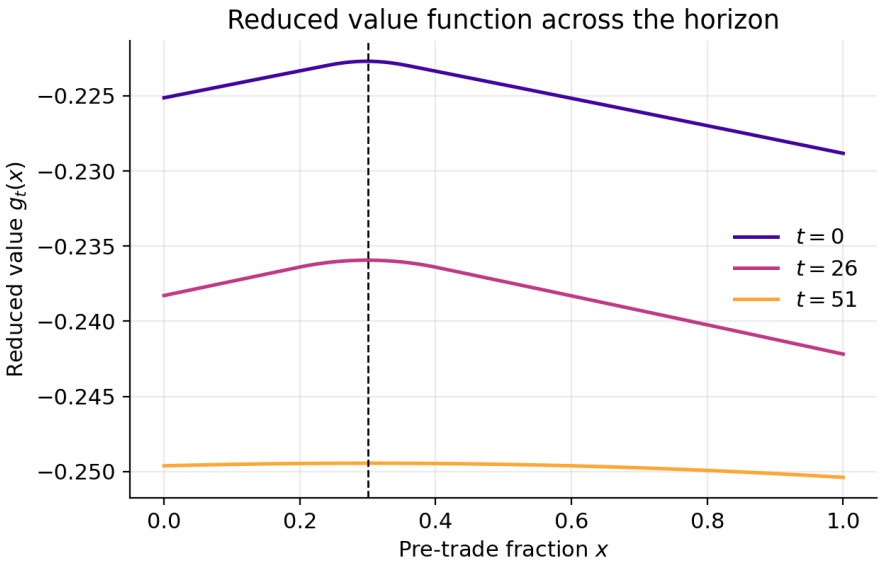


Figure 1: The reduced value function $g_t(x)$ at $t = 0$, $t = 26$ and $t = 51$ (dashed line: the Merton fraction $w^* = 0.30$). Each slice peaks at w^*, and the curve flattens as the terminal date approaches — the flatter g_t, the less a misallocation costs, and the wider the no-trade band.

Data provenance. We state plainly what these numbers are and are not. *This study uses no empirical data.* Every return path in the paper is drawn from the model of Equation (1) under a single seed; there is no historical price series, no external dataset, and no estimation step anywhere in the pipeline. The parameters of Table 1 are *stylized and illustrative*, not calibrated: they are round numbers of the right order for a broad equity index over a one-year horizon (8% drift, 20% volatility, 2% risk-free rate), chosen so that w^* sits in the interior of the admissible interval and the no-trade band is legible. The cost rate $tc = 1\%$ is likewise a round figure, an order of magnitude above liquid equity execution costs, chosen to make the band wide enough to see; Section 6 reports the full comparative static in tc precisely so that no conclusion rests on this choice.

The consequence for interpretation is worth being explicit about. Every result below is a statement about *this model*, not an empirical claim about any market. What we take to be transportable is the qualitative structure — the existence of the no-trade region, the direction of its comparative statics, and the failure mode of action-space imitation — all of which are properties of the control problem rather than of the particular numbers in Table 1. The point estimates (a band of $[0.235, 0.365]$, a certainty-equivalent gap of 3×10^{-3}) are not.

5 Machine-Learning Approximation

5.1 Dataset generation

We treat the dynamic-programming policy as a teacher and form the supervised dataset $\{((\tau_t, x_j),\, y^*(t, x_j))\}$ over all dates and grid points, where $\tau_t = t/N \in [0, 1]$ is normalized time-to-go and x_j the pre-trade fraction. This yields $N \times n_x = 52 \times 201 = 10{,}452$ examples.

Table 1: Base-case parameters.

Symbol	Meaning	Value
μ	risky drift (annual)	0.08
σ	risky volatility (annual)	0.20
r	risk-free rate (annual)	0.02
γ	relative risk aversion	5.0
tc	proportional cost rate	0.01
T	horizon (years)	1.0
N	rebalancing dates	52 (weekly)
n_x	allocation grid points	201
n_q	quadrature nodes	21
w^*	Merton fraction	0.300

5.2 Network architecture

The student is a multilayer perceptron mapping $(\tau, x) \mapsto y$ with two hidden layers of 64 units and tanh activations, and a sigmoid output so predictions lie in $[0, 1]$ by construction. The network has 4,417 parameters — deliberately small, to test ML as a *lightweight* surrogate.

5.3 Training methodology

We minimize mean-squared error with Adam [Kingma and Ba, 2015] (learning rate 10^{-3}, weight decay 10^{-5}, batch size 256, 400 epochs), holding out 15% of examples for validation. Training converges to a validation MSE of 2.1×10^{-4} in under 20 s on CPU (Figure 2). Inference is a single forward pass — no grid lookup or interpolation — so the learned controller is orders of magnitude cheaper to query than the tabulated optimal policy.

Figure 2 shows the optimization is entirely well behaved: training and validation curves fall together over three orders of magnitude and stay locked to each other, with no separation at any point. The student neither underfits nor overfits its teacher. We stress this because it rules out the obvious explanations for the failure documented in Section 6. The network's poor economic performance is not a capacity limit, an optimization failure, or a generalization gap — by every diagnostic available in action space, the imitation succeeded.

6 Experimental Results

6.1 No-trade regions

Figure 3a shows the optimal no-trade region across the horizon. For most of the horizon it is a narrow, nearly constant band around $w^* = 0.30$; at $t = 0$ it is $[0.235,\ 0.365]$ (width 0.130). Near the terminal date the band flares open — the *horizon effect* — because with little time remaining there is too little expected benefit from re-optimizing the allocation to justify paying the cost. Figure 3b displays the policy slices $y^*(x)$, exhibiting the bang-bang-to-boundary structure of (8): flat at the lower edge, the identity $y = x$ inside the band, flat at the upper edge.

Comparative statics. Tables 2 and 3 (and Figure 4) report how the $t = 0$ band responds to the cost rate and to risk aversion. The band widens monotonically with tc — from width 0.040 at

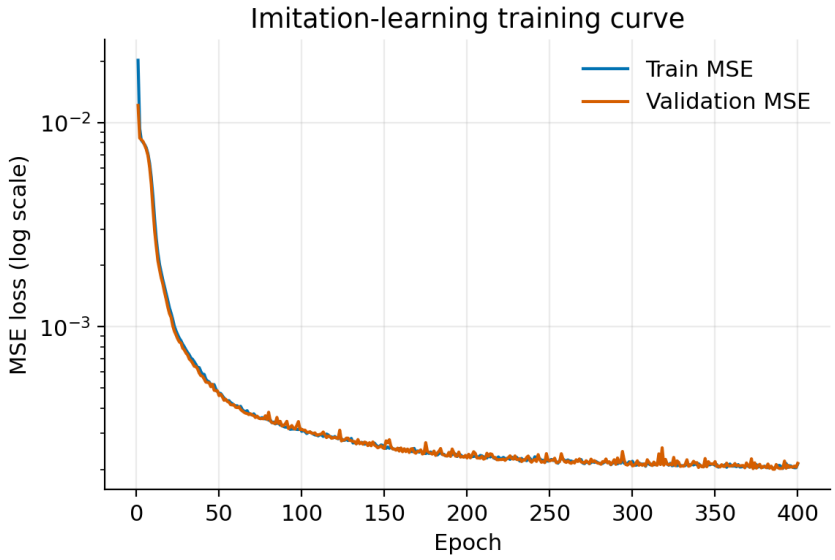


Figure 2: Imitation-learning training curve (log scale). Train and validation MSE descend together to $\approx 2.1 \times 10^{-4}$ with no divergence: the network fits the optimal policy well in action space. Its economic failure (Section 6) is therefore structural, not a training pathology.

Table 2: No-trade band at t=0 vs cost.

tc	lower	upper	width
0.0005	0.280	0.320	0.040
0.001	0.275	0.325	0.050
0.002	0.270	0.330	0.060
0.005	0.255	0.345	0.090
0.01	0.235	0.365	0.130
0.02	0.190	0.415	0.225
0.03	0.140	0.470	0.330

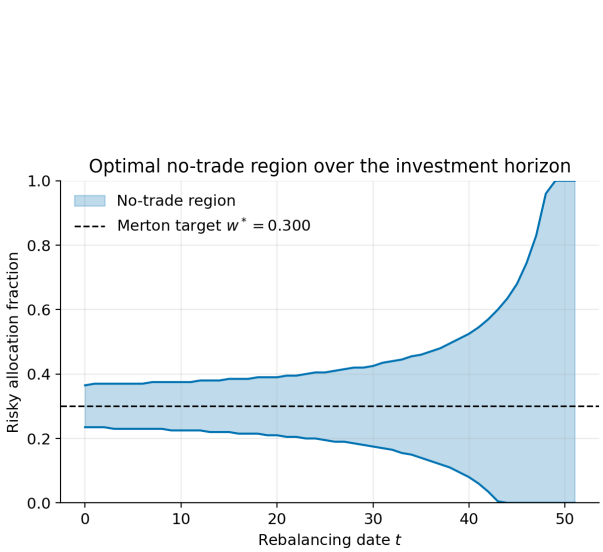
Table 3: Band at t=0 vs risk aversion.

γ	w^*	lower	upper	width
2	0.750	0.615	0.885	0.270
3	0.500	0.400	0.600	0.200
5	0.300	0.235	0.365	0.130
8	0.188	0.140	0.235	0.095
12	0.125	0.090	0.160	0.070

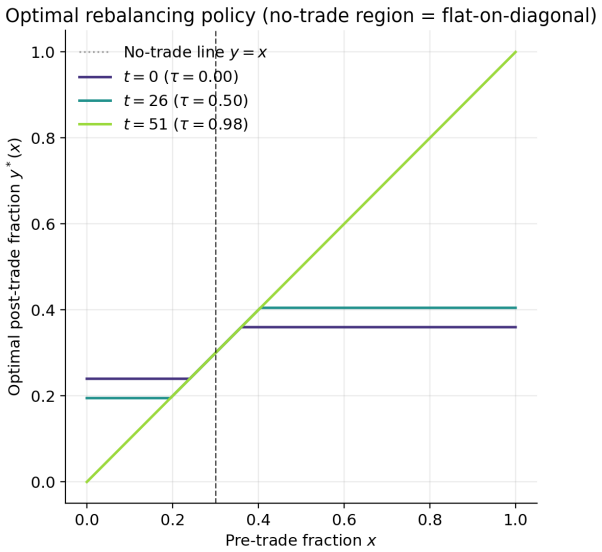
$tc = 0.05\%$ to 0.330 at $tc = 3\%$. For small costs the half-width roughly triples when tc increases sixteen-fold ($0.0005 \to 0.002$ gives $0.020 \to 0.030$, ratio $1.5 \approx 4^{1/3}$), consistent with the classical $tc^{1/3}$ small-cost asymptotics [Shreve and Soner, 1994]. Raising γ lowers the target w^* and tightens the band (width 0.270 at $\gamma = 2$ down to 0.070 at $\gamma = 12$): more risk-averse investors hold less of the risky asset and tolerate smaller deviations.

6.2 Policy comparisons: learned vs optimal

Figure 5 overlays the learned policy on the optimal one. The network captures the global shape — low (0.235-like) lower plateau, diagonal no-trade segment, upper plateau — but *smooths the kinks*. Crucially, inside the no-trade band where the optimum is exactly $y = x$, the smooth approximation lies slightly off the diagonal. Measured inside the $t = 0$ band, the mean residual $|y_{\mathrm{ML}} - x|$ is small but strictly positive everywhere. Each such residual is a trade the optimal policy would not make.



(a) No-trade region over time.



(b) Optimal policy $y^*(x)$ at three dates.

Figure 3: The optimal control. The band (left) is centered on the Merton target and widens toward the horizon; the policy (right) holds inside the band and trades to its nearest edge outside.

Table 4: Performance starting *at* the target ($x_0 = w^* = 0.30$), 20,000 paths. CE shown with 95% confidence interval.

Strategy	CE	95% CI	$\mathbb{E}[W_T]$	Turnover	Cost	Track. err.
DP optimal	1.02927	[1.02842, 1.03013]	1.0390	0.00003	4×10^{-7}	—
Buy & hold	1.02927	[1.02842, 1.03013]	1.0390	0.00000	0	0.0002
Periodic	1.02821	[1.02735, 1.02908]	1.0375	0.1119	0.00114	0.0056
ML imitation	1.02606	[1.02500, 1.02713]	1.0402	0.2141	0.00217	0.0052

6.3 Performance metrics

We simulate all four strategies — DP optimal, ML imitation, periodic rebalancing (to w^* every 4 steps), and buy-and-hold — on a common set of 20,000 return paths (common random numbers). Performance is summarized by the certainty equivalent $\text{CE} = \big((1-\gamma)\,\mathbb{E}[U(W_T)]\big)^{1/(1-\gamma)}$, mean turnover, mean cost, and the root-mean-square action gap to the optimal policy (tracking error). Because all strategies see identical returns, we also report a *paired* test of the per-path utility advantage of the optimal policy, whose variance is far smaller than that of the individual estimates.

Three findings emerge (Tables 4–6, Figure 6).

1. **The DP policy is optimal, as it must be.** It attains the highest certainty equivalent and dominates ML and periodic rebalancing with large t-statistics (20–29). When the portfolio starts *on* target it ties buy-and-hold ($t = -0.4$, not significant): a wide no-trade band makes optimal behavior essentially "do nothing," so the do-nothing heuristic is optimal here. Starting *off* target, the optimal policy beats every baseline including buy-and-hold, trading in most cost-efficiently (turnover 0.45 vs 0.50 for buy-and-hold and 0.61 for periodic).

2. **Naive imitation is the worst strategy.** Despite a tiny validation MSE and a small tracking error (0.005), the ML policy has by far the highest turnover (0.21 vs 3×10^{-5} for DP) and

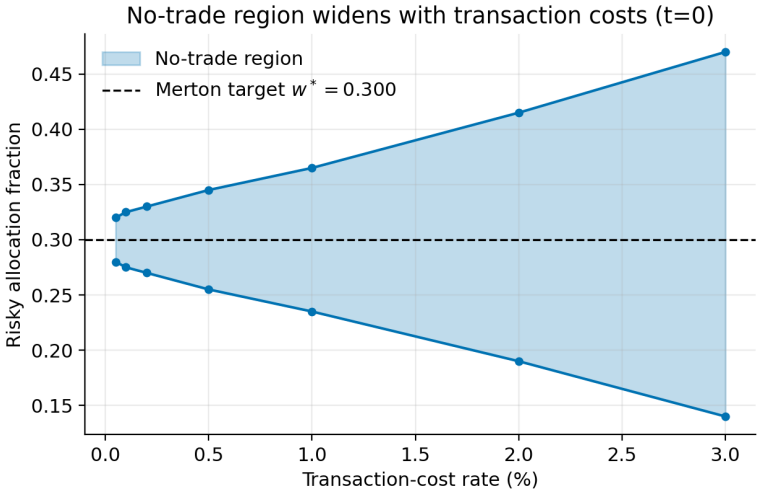


Figure 4: The no-trade region widens with the transaction-cost rate, centered on the Merton target.

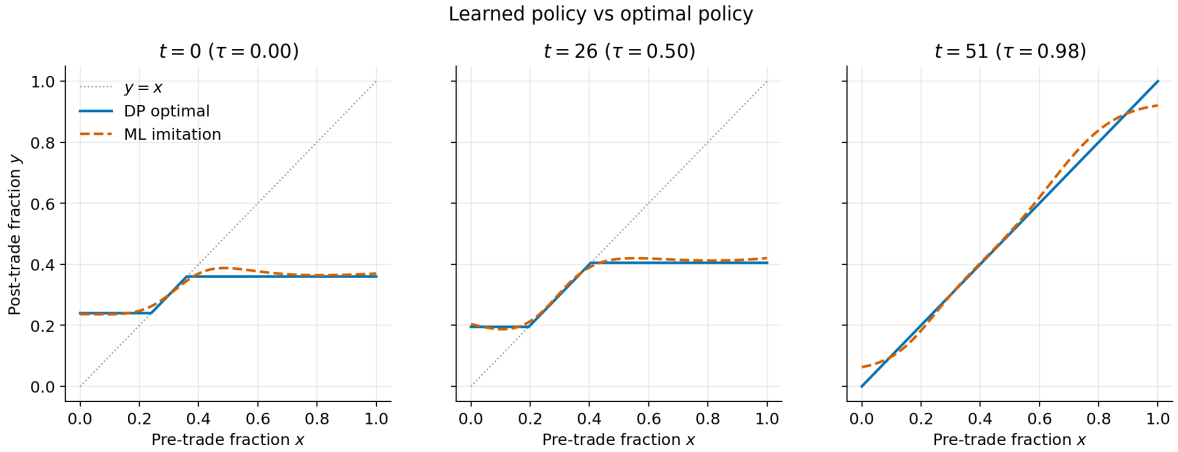


Figure 5: Learned (ML) vs optimal (DP) policy at three dates. The network smooths the no-trade kinks; the residual off-diagonal inside the band drives spurious trading.

cost, and the lowest certainty equivalent. The smooth network cannot reproduce the exact $y = x$ identity on the positive-measure no-trade region, so it trades *continuously* where the optimum holds.

3. **Low action error $\neq$ good control.** The disconnect between ML's low tracking error and poor economic performance is the paper's central lesson: errors inside the inaction region are economically amplified into turnover and cost, a structure invisible to an action-space MSE objective.

6.4 Repairing the imitation

A diagnosis is worth little unless it predicts a fix. If the failure really is the missing inaction region — and not capacity, optimization, or generalization — then restoring that region should recover the lost performance, and nothing else should be required. We test two ways of doing so.

Table 5: Performance starting *off* the target ($x_0 = 0.80$), 20,000 paths.

Strategy	CE	95% CI	$\mathbb{E}[W_T]$	Turnover	Cost	Track. err.
DP optimal	1.02393	[1.02293, 1.02494]	1.0372	0.4526	0.00453	—
Buy & hold	1.02378	[1.02292, 1.02464]	1.0337	0.5000	0.00500	0.0083
Periodic	1.02269	[1.02183, 1.02356]	1.0320	0.6120	0.00614	0.0100
ML imitation	1.02063	[1.01956, 1.02171]	1.0351	0.7065	0.00709	0.0065

Table 6: Paired utility advantage of the DP-optimal policy (common random numbers). Positive means DP is better; t-statistics in the final column.

Start	DP vs	Mean adv. ($\times 10^3$)	t-stat	DP better @ 95%
on-target	ML imitation	+2.799	20.1	yes
on-target	Periodic	+0.918	24.1	yes
on-target	Buy & hold	−0.0001	−0.4	no (tie)
off-target	ML imitation	+2.959	28.8	yes
off-target	Periodic	+1.108	17.0	yes
off-target	Buy & hold	+0.140	2.0	yes

A dead-zone wrapper (inference time). Leave the trained network untouched and snap small predicted trades to exact inaction:

$$y = \begin{cases} x, & |\hat{y}(t,x) - x| \le \delta, \\ \hat{y}(t,x), & \text{otherwise.} \end{cases} \tag{9}$$

Returning x *exactly* is what matters: cost is charged on any deviation, so only an exact match is genuinely free. This restores an inaction region of half-width δ by construction, costs nothing to train, and adds one comparison per step.

A cost-aware objective (training time). Alternatively, charge the student during training for trading more than its teacher does:

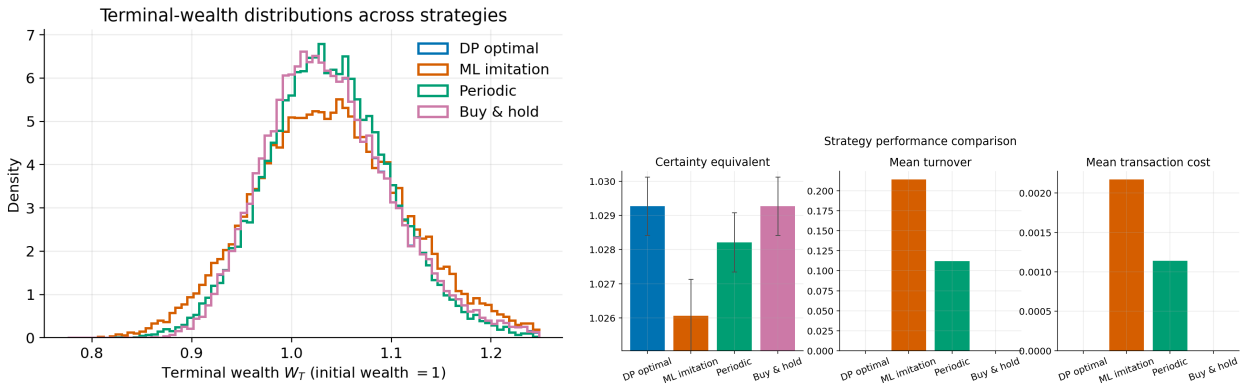
$$\mathcal{L} = \mathrm{MSE}(\hat{y}, y^*) + \lambda\, \mathbb{E}\big[(\,|\hat{y} - x| - |y^* - x|\,)^+\big]. \tag{10}$$

The asymmetry is deliberate. Penalizing $|\hat{y} - x|$ outright would also discourage the large, correct trades made from outside the band, biasing the student toward inaction everywhere. Penalizing only the *excess* over the teacher's own trade size leaves correct trading unpunished and bills the student exactly for the spurious turnover it invents inside the band. We use $\lambda = 1$.

Table 7 and Figure 7 report the outcome, and it is unambiguous.

1. **The dead-zone recovers the optimum exactly.** At $\delta = 0.05$ the learned policy attains a certainty equivalent of 1.029272 — the DP value to six decimals — with turnover of 9×10^{-17}, i.e. it stops trading altogether when it should. The paired test against the optimum gives $t = -0.4$: after the repair, the network is *no longer statistically distinguishable from the optimal control*, having been beaten at $t = 20.1$ before it.

2. **The transition is where the mechanism says it should be.** The network's mean in-band residual is 9.6×10^{-3}. Dead-zones below that value do essentially nothing (3.5% of the gap at



(a) Terminal-wealth distributions.

(b) CE (with 95% CI), turnover, cost.

Figure 6: Strategy comparison on a common set of paths (on-target start).

Table 7: Repairing the learned policy, on-target start, 20,000 paths and the same return paths as Table 4. "Gap closed" is the fraction of the DP–ML certainty-equivalent gap (3.208×10^{-3}) recovered. The final column is the paired t-statistic against the DP optimum; small values mean the policy is no longer statistically distinguishable from optimal.

Policy	CE	Turnover	Gap closed	t vs DP
DP optimal	1.029272	3.4×10^{-5}	—	—
ML imitation	1.026064	0.2141	0%	20.1
+ dead-zone $\delta = 0.002$	1.026175	0.2032	3.5%	19.4
+ dead-zone $\delta = 0.005$	1.026602	0.1714	16.8%	18.4
+ dead-zone $\delta = 0.01$	1.027331	0.1277	39.5%	18.4
+ dead-zone $\delta = 0.02$	1.029146	0.0083	96.1%	7.7
+ dead-zone $\delta = 0.05$	**1.029272**	9×10^{-17}	**100**%	**−0.4**
ML cost-aware	1.027807	0.1065	54.3%	17.4
+ dead-zone $\delta = 0.01$	**1.029272**	4.8×10^{-5}	**100**%	2.1

$\delta = 0.002$); the recovery happens as δ crosses it (39.5% at $\delta = 0.01$, 96.1% at $\delta = 0.02$, 100% at $\delta = 0.05$). The fix works exactly when it covers the defect it was designed for, which is the strongest evidence we can offer that the diagnosis is the right one.

3. **The cost-aware loss attacks the same quantity, from the other side.** It shrinks the in-band residual from 9.6×10^{-3} to 2.6×10^{-3}, a factor of 3.8, and closes 54.3% of the gap on its own with no wrapper at all. Combined, the two are complementary: because the residual is smaller, a dead-zone *five times tighter* ($\delta = 0.01$) suffices to reach the optimum.

4. **δ is a real trade-off, not a free parameter.** A dead-zone wide enough to cover the residual also suppresses genuine rebalancing. Starting off-target the same $\delta = 0.05$ closes 99.2% rather than 100% of the gap, and under-trades relative to the optimum (turnover 0.438 versus 0.453); pushed far enough, a dead-zone simply refuses to correct a bad allocation. This is why the cost-aware loss matters: it buys the same repair at a tighter, safer δ.

The practical reading is that neither remedy is exotic — one is three lines at inference, the other one term in the loss — and either is enough to turn the worst strategy in Table 4 into one that ties

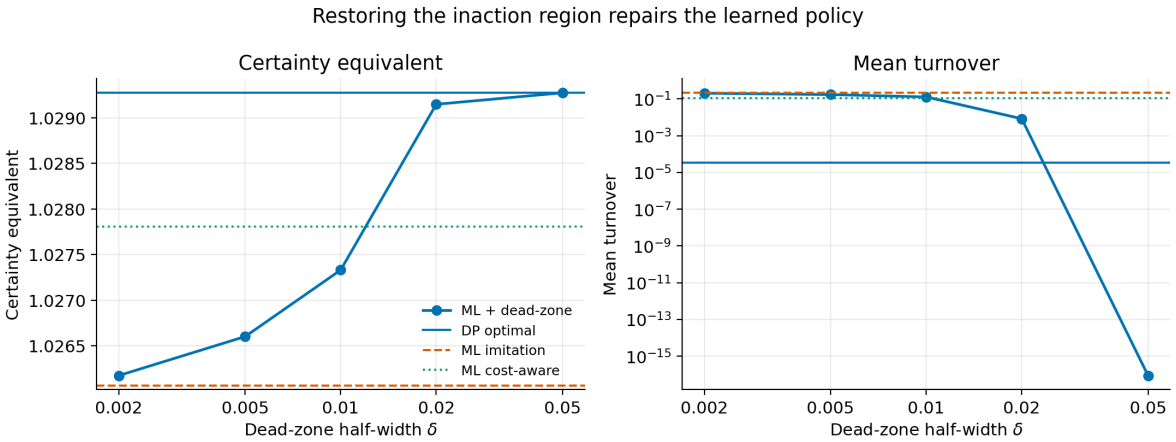


Figure 7: Certainty equivalent and turnover of the dead-zone-wrapped policy as δ varies, against the DP optimum and the two ML endpoints. The transition occurs where δ crosses the network's in-band residual.

the optimum. What they have in common is that both encode the inaction structure rather than hoping the regressor will discover it.

6.5 Is the failure robust, or is it one unlucky fit?

Everything so far rests on a single trained network. That is enough to *describe* a failure but not to establish one: the obvious objection is that we happened to train a bad student. We therefore repeat the imitation experiment across 10 independent training seeds at the published architecture, and across 7 architectures at the base seed, spanning 337 to 66,817 parameters — a factor of 198 in capacity. All 17 students are evaluated on the same 20,000 paths as Table 4, against the same teacher, which is seed-independent and solved once.

Across seeds. Every one of the 10 retrainings fits well and behaves badly. Validation MSE stays in $[1.65, 2.50]\times10^{-4}$; the in-band residual is $1.10\times10^{-2}\pm1.4\times10^{-3}$; realized turnover is 0.236 ± 0.039 against the optimal policy's 3.4×10^{-5}; and the certainty equivalent lies in $[1.02437, 1.02662]$, *below the DP optimum of* 1.02927 *in* 10 *cases out of* 10. There is no seed at which naive imitation is competitive. The mean certainty-equivalent shortfall is 3.41×10^{-3}, close to the 3.21×10^{-3} of the headline run, so the paper's single-seed result is typical rather than extreme.

Across architectures. Table 8 makes the sharper point. Capacity does not help, and the two natural "just train a better network" responses both fail.

1. **More parameters do not restore inaction.** The 256×256 network has 198 times the parameters of the 16×16 network and a *larger* in-band residual (0.01662 versus 0.01043, a factor of 1.59 in the wrong direction). Across the whole sweep the residual never falls below 7.5×10^{-3} and shows no downward trend in capacity. This is what the representational argument predicts: the obstruction is that a smooth function cannot equal the identity on a positive-measure set, and no amount of width changes that.

2. **Fitting better does not control better.** The best fit in the entire study is the four-layer network, whose validation MSE of 8.92×10^{-5} is 2.4 times smaller than the published run's.

Table 8: The failure across architectures (base seed, 20,000 paths). Capacity spans a factor of 198 and depth ranges from two to four hidden layers. Every row under-performs the DP optimum (CE = 1.02927); every row repairs to it exactly under a $\delta = 0.05$ dead-zone.

Architecture	Params	Val. MSE	In-band $\lvert\hat{y} - x\rvert$	Turnover	CE
16×16	337	2.18×10^{-4}	0.01043	0.2596	1.02532
32×32	1,185	2.34×10^{-4}	0.00903	0.2750	1.02478
64×64	4,417	2.14×10^{-4}	0.00961	0.2141	1.02606
128×128	17,025	2.01×10^{-4}	0.01251	0.2019	1.02632
256×256	66,817	2.08×10^{-4}	**0.01662**	0.2668	1.02544
$64 \times 64 \times 64$	8,577	9.83×10^{-5}	0.00751	0.1806	1.02672
$64 \times 64 \times 64 \times 64$	12,737	$\mathbf{8.92 \times 10^{-5}}$	0.00799	0.2147	1.02649
DP optimal	—	—	0 (exactly)	0.00003	**1.02927**

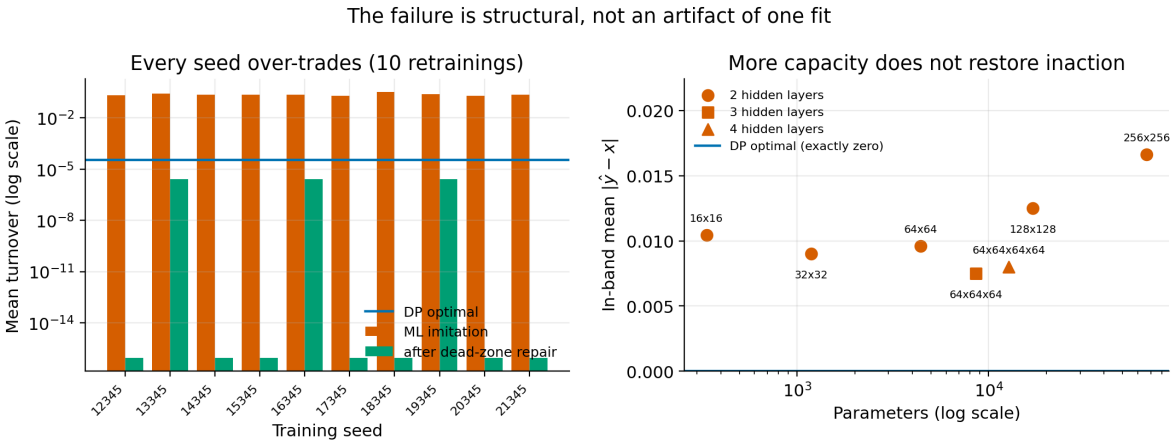


Figure 8: Left: realized turnover for each of the 10 retraining seeds, before and after the dead-zone repair, against the optimal policy. Right: in-band residual against parameter count; the horizontal line at zero is the optimal policy, which the students never approach at any capacity.

> It still turns over 0.215 and still loses to the optimum. Ranking the seventeen students by validation error tells you essentially nothing about their economics — which is the paper's central lesson, now stated over a population of fits rather than a single one.

The repair is equally universal. Applying the $\delta = 0.05$ dead-zone of Equation (9) to each of the 17 students, every one attains a certainty equivalent of 1.02927 — the DP value, to five decimals, in all seventeen cases — with residual turnover of at most 2.6×10^{-6}. The repair is not tuned per fit and does not depend on the architecture or the seed. Both the disease and the cure are properties of the control problem, not of any particular network.

7 Discussion

Interpretation of the learned policy. The failure mode is not a capacity or optimization problem — the network fits the policy well in MSE, and Section 6.5 shows that neither more parameters nor a better fit rescues it — but a *representational mismatch* with the geometry of optimal control. Optimal rebalancing is governed by a free boundary (the band edges) and an

identity map in between. A smooth regressor approximates the identity map with a wiggle, and because the wealth dynamics charge cost on *any* deviation, a wiggle of size ε produces turnover of order ε per step, accumulating to first order in N while the utility benefit of being marginally better allocated is only second order. Hence small ε still hurts.

Remedies. Two cheap fixes follow directly from this diagnosis, and Section 6.4 shows both work: a *dead-zone* wrapper that snaps small predicted trades to exact inaction, and a *cost-aware loss* that charges the student for trading beyond its teacher. Either closes the entire certainty-equivalent gap, and the dead-zone begins to work exactly where δ crosses the network's in-band residual — the fix engages precisely when it covers the defect, which is the sharpest confirmation of the mechanism we can give. A third route we have not tested is imitation in *value* rather than action space (regress g_t and act greedily), which would sidestep the representational problem entirely by never asking a smooth function to reproduce a kink.

Computational trade-offs. For this one-dimensional problem the DP solve is already trivial ($\approx 0.02\,\mathrm{s}$), so ML offers no speed advantage here; its appeal is asymptotic, in higher-dimensional settings where the DP grid is infeasible. Our result is a cautionary calibration: before scaling imitation to many assets, the inaction structure must be built into the hypothesis class or the loss.

Practical implications. The strong showing of buy-and-hold on a one-year horizon, and of the optimal policy's near-inaction when started on target, both underline that *the cheapest correct action is usually to do nothing.* Turnover is a cost, not a virtue; periodic rebalancing pays for trades the optimal policy would skip.

8 Limitations and Future Work

Our model is deliberately minimal. (i) **Multi-asset extension**: the homogeneity reduction survives with multiple risky assets, but the no-trade region becomes a multi-dimensional polytope and the grid grows exponentially — precisely where learned controllers should help [Liu, 2004]. (ii) **Stochastic volatility** and predictable returns [Gârleanu and Pedersen, 2013] would make the band time- and state-dependent in richer ways. (iii) **Deep variational solvers**: rather than imitating a precomputed policy, one can solve the Hamilton–Jacobi–Bellman equation directly with neural networks [Han et al., 2018] or train under a frictional P&L objective as in deep hedging [Buehler et al., 2019], sidestepping the teacher entirely. (iv) We use proportional costs only; fixed or quadratic costs change the geometry (e.g. fixed costs induce discrete jumps to an interior target). (v) Intermediate consumption and an infinite horizon are natural extensions of the present terminal-wealth formulation.

(vi) The robustness study of Section 6.5 varies the seed and the architecture but not the *model class*: every student is a feedforward MLP trained by regression on the policy. Architectures with built-in discontinuities (ReLU networks with learned thresholds, mixture-of-experts gating, or an explicit hold/trade classification head) might represent the inaction region natively, and our result should not be read as a claim about all learned controllers — only about smooth action-space regression, which is the default choice.

Above all, (vii) **there is no empirical component**. Returns are simulated from the assumed geometric Brownian motion and the parameters are stylized rather than estimated, so the paper offers no evidence that the bands reported here are the right bands for any actual portfolio. Real returns are not lognormal — they are fat-tailed, volatility-clustered and serially dependent — and

real execution costs are neither purely proportional nor constant across trade sizes. Confronting the solver with historical prices, estimating μ and σ with their sampling error propagated into the band, and measuring realized turnover against a live cost model are all necessary before any of the numbers here could guide an allocation decision. We regard that as the natural next step rather than a defect of the present exercise, whose aim is to establish the structure of the control problem and the failure mode of naive imitation.

9 Conclusion

We formulated portfolio rebalancing under proportional transaction costs as a finite-horizon stochastic control problem, reduced it to one dimension via CRRA homogeneity, and solved it exactly by backward induction. The solver reproduces the classical no-trade region and its comparative statics — widening with cost ($\sim tc^{1/3}$), narrowing with risk aversion, and flaring near the horizon. Testing a lightweight neural network as a fast approximate controller, we found that naive supervised imitation, though accurate in mean-squared action error, is the worst-performing strategy because it cannot represent the no-trade region and therefore over-trades. Crucially, the diagnosis proved actionable: encoding the inaction region — either as a dead-zone wrapper at inference or as an excess-turnover penalty in the loss — closed the entire certainty-equivalent gap and left the repaired policy statistically indistinguishable from the optimal control, with the recovery setting in exactly where the dead-zone width crosses the network's in-band residual. The lesson generalizes beyond this model: imitating optimal control requires encoding the inaction structure, not merely minimizing action error, and doing so is cheap once the failure is understood. All figures, tables, and checkpoints are reproducible from the accompanying code via a single command.

Reproducibility and code availability

Every number, table and figure in this paper is regenerated by two commands:

- `python scripts/run_all.py` — the solver, the imitation study and the remedies of Section 6.4; about 40 seconds on a laptop CPU.
- `python scripts/run_robustness.py` — the seed and architecture sweep of Section 6.5, which retrains the network seventeen times; about seven minutes.

All randomness derives from one seed (`Config.seed` = 12345) and all economic and numerical parameters live in a single `Config` dataclass, so the run is deterministic: repeated executions reproduce the committed tables, checkpoints and figures *byte for byte*. The only artifacts excluded from that guarantee are wall-clock timings, which are machine-dependent and are written to a separate file rather than mixed into the metrics.

The repository ships an automated test suite (`pytest`) that checks the claims made here rather than merely exercising the code. It verifies the quadrature against the analytic lognormal moments; the proportional-cost identity of Proposition 1, including that no-trade is exactly costless; that the no-trade band brackets the Merton fraction, widens monotonically with tc at a fitted log–log slope consistent with the $tc^{1/3}$ asymptotics, narrows with γ, and collapses to the Merton point when $tc = 0$; that the band is stable under grid and quadrature refinement; that the learned policy is off the $y = x$ diagonal at *every* point inside the band, which is the mechanism of Section 6; and that every figure, table entry and quoted statistic in this manuscript agrees with the generated results file. The environment the reported numbers were produced in (Python 3.10.11, NumPy 1.26.4, SciPy 1.15.3, PyTorch 2.7.1, CPU only) is pinned in `requirements-lock.txt`.

Declarations

Code availability. All code needed to regenerate every figure, table and number in this paper is openly available at `https://github.com/Raahul-Thakur/optimal-portfolio-rebalancing`, archived at [ARCHIVED DOI TO BE INSERTED]. The repository contains the solver, the training and evaluation pipeline, the manuscript source, and an automated test suite; `python scripts/run_all.py` reproduces the complete set of results, and `pytest` verifies both the numerical claims and the agreement between this manuscript and the generated results.

Data availability. This study generated and analysed no empirical data. Every return path is simulated from the model of Equation (1) under a single fixed seed, as described in Section 4; the simulated data are therefore fully specified by the parameters of Table 1 and are regenerated deterministically by the code above rather than distributed as files.

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